Shopify Technical Challenge Submission

Tolu Fagorusi

Question 1: Given some sample data, write a program to answer the following: click here to access the required data set

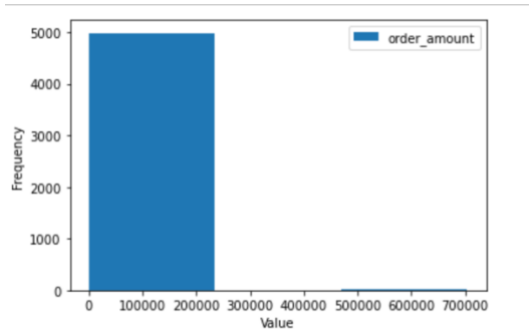
> On Shopify, we have exactly 100 sneaker shops, and each of these shops sells only one model of shoe. We want to do some analysis of the average order value (AOV). When we look at orders data over a 30 day window, we naively calculate an AOV of $3145.13. Given that we know these shops are selling sneakers, a relatively affordable item, something seems wrong with our analysis.

a. Think about what could be going wrong with our calculation. Think about a better way to evaluate this data.
b. What metric would you report for this dataset?
c. What is its value?

Solution:

a. First by using boxplot to show the distribution of the 'order-amount' column in the dataset, it can be observed that the calculated AOV of $3145.13 is likely to be incorrect due to the influence of the outliers as shown in Fig 1a. A better way to evaluate this dataset would be to remove the outlier values.

Box plot



Histogram

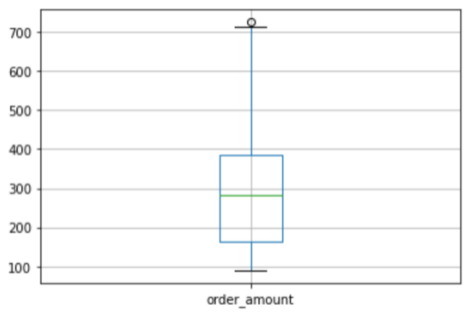
Figure 1a

The quartile range method was used to obtain a threshold needed to remove outliers and obtain a 'new' dataset using the python script shown below

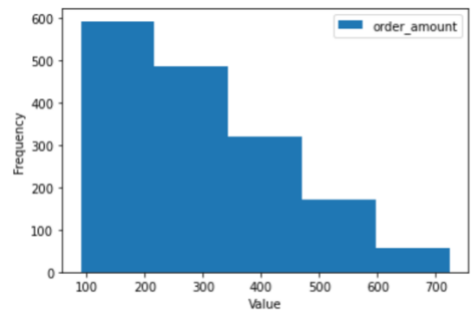
```python
import numpy as np
import statistics as st


def remove_outliers(df, col):
    # remove outliers
    q25, q75 = np.percentile(df[col], 25), np.percentile(df[col], 75) #calculate interquartile range
    iqr = q75 - q25
    # calculate the outlier cutoff
    cut_off = iqr * 1.5
    lower, upper = q25 - cut_off, q75 + cut_off
    df_filtered = df[(df[col] < upper) & (df[col] > lower)]
    return df_filtered
```

Figure 1b

The boxplot and the histogram plot of the 'order_amount' column in the 'new' dataset is given below



Box plot



Histogram

Figure 1c

b. As observed from the histogram plot above, it can be observed that the data is skewed (that is, the frequency distribution of the data is skewed). In this case, the Median is preferred as a metric to report this dataset.

c. Based on the 'new' data (df_filtered), the value obtained is $284. In comparison with the 'original' dataset (data), using the suggested Median metric would give $280 which is much better than the previously reported AOV of $3145.13 using the mean metric.

```python
st.median(df_filtered['order_amount'])
```

```
280
```

```python
st.median(data['order_amount'])
```

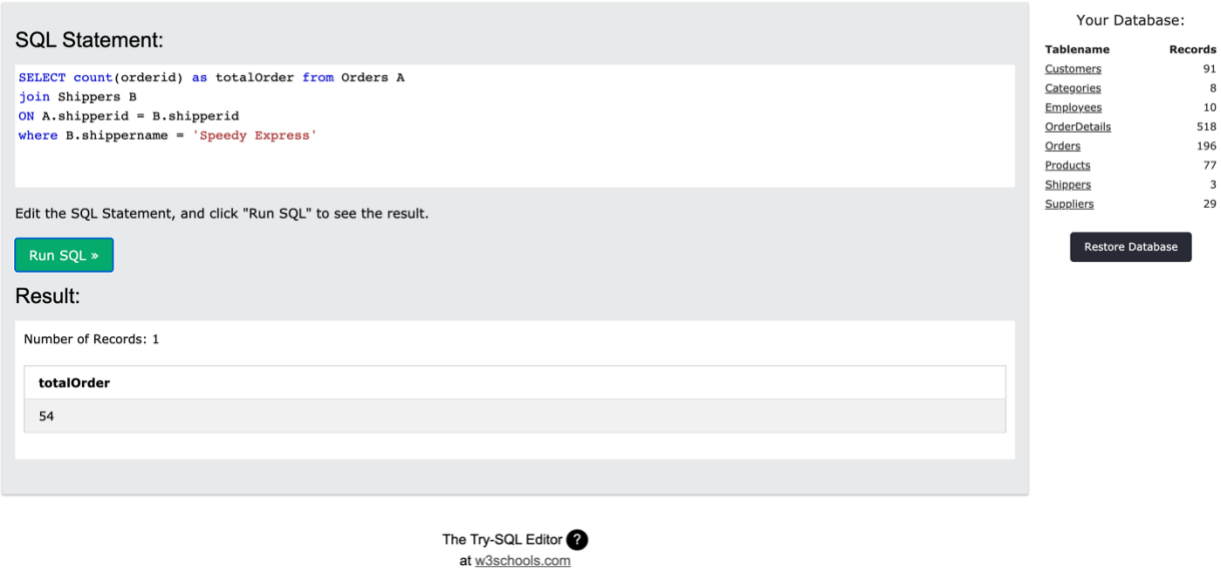
```
284.0
```

Question 2: For this question you'll need to use SQL. Follow this link to access the data set required for the challenge. Please use queries to answer the following questions. Paste your queries along with your final numerical answers below.

Solution

a. How many orders were shipped by Speedy Express in total?

SELECT count(orderid) as totalOrder from Orders A
join Shippers B
ON A.shipperid = B.shipperid
where B.shippername = 'Speedy Express'



The Try-SQL Editor
at w3schools.com

Figure 2a

b. What is the last name of the employee with the most orders?

SELECT myTable.lastname from
(SELECT employees.lastname, count(distinct orders.orderid)as totalOrder from orders
join employees
on orders.employeeid = employees.employeeid
group by employees.lastname
order by totalOrder desc) myTable
limit 1

SQL Statement:

```
SELECT myTable.lastname from
(SELECT employees.lastname, count(distinct orders.orderid)as totalOrder from orders
join employees
on orders.employeeid = employees.employeeid
group by employees.lastname
order by totalOrder desc) myTable
limit 1
```

Edit the SQL Statement, and click "Run SQL" to see the result.

Run SQL »

Result:

Number of Records: 1

lastname
Peacock

Your Database:

Tablename	Records
Customers	91
Categories	8
Employees	10
OrderDetails	518
Orders	196
Products	77
Shippers	3
Suppliers	29

Restore Database

The Try-SQL Editor
at w3schools.com

Figure 2b

c. What product was ordered the most by customers in Germany?

SELECT productname
from orderdetails A
join products B on A.productid = B.productid
join orders C on A.orderid = C.orderid
join customers D on C.customerid = D.customerid
group by productname
having country = 'Germany'
order by quantity desc
limit 1

SQL Statement:

```
SELECT productname
from orderdetails A
join products B on A.productid = B.productid
join orders C on A.orderid = C.orderid
join customers D on C.customerid = D.customerid
group by productname
```

Edit the SQL Statement, and click "Run SQL" to see the result.

Run SQL »

Result:

Number of Records: 1

ProductName
Boston Crab Meat

Your Database:

Tablename	Records
Customers	91
Categories	8
Employees	10
OrderDetails	518
Orders	196
Products	77
Shippers	3
Suppliers	29

Restore Database

The Try-SQL Editor
at w3schools.com

Figure 2c